multiple operating systems and software platforms, which can be directly pulled and used in it.

New generation virtualisation using Docker

DevOps and rapid applications development (RAD) using advanced virtualisation platforms are immensely popular today. Docker (*https://www.docker.com/*), a high performance platform used for containerisation and rapid applications development, enables the apps to be developed quickly, and shared and deployed anywhere.

Working with Docker on different platforms

Docker is a cross-platform and open source application that can be deployed on multiple operating systems without any issues of compatibility and performance. Its variants are available for Windows, Mac, Linux and cloud platforms.

Docker has a strong base of developers and developed applications. More than 13 million developers, 13 billion monthly image downloads, and 7 million applications are associated with the Docker platform. The prominent corporate users of Docker include Adobe, Anaplan, Blue Apron, PayPal, Segment, Stripe, Verizon, Yale, University of Calgary, Splunk, PathFactory, Paloalto, Netflix, AT&T, and Lucent Health.

Docker deployment

Docker comes in various forms.

- *Docker for Windows:* Available for Windows 10 or above, as the support for Windows 7 and Windows 8 has been withdrawn
- *Docker for MacOS*
- *Docker for Linux:* Installation on Linux kernel version 3.8 or higher
- *Docker Engine:* For building images and creating containers
- *Docker Hub:* Public repository and registry to host images
- *Docker Compose:* Use of multiple Docker containers

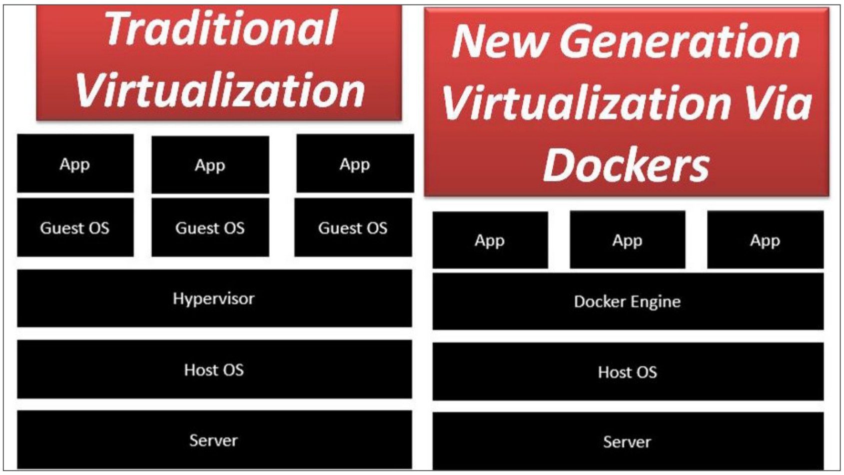
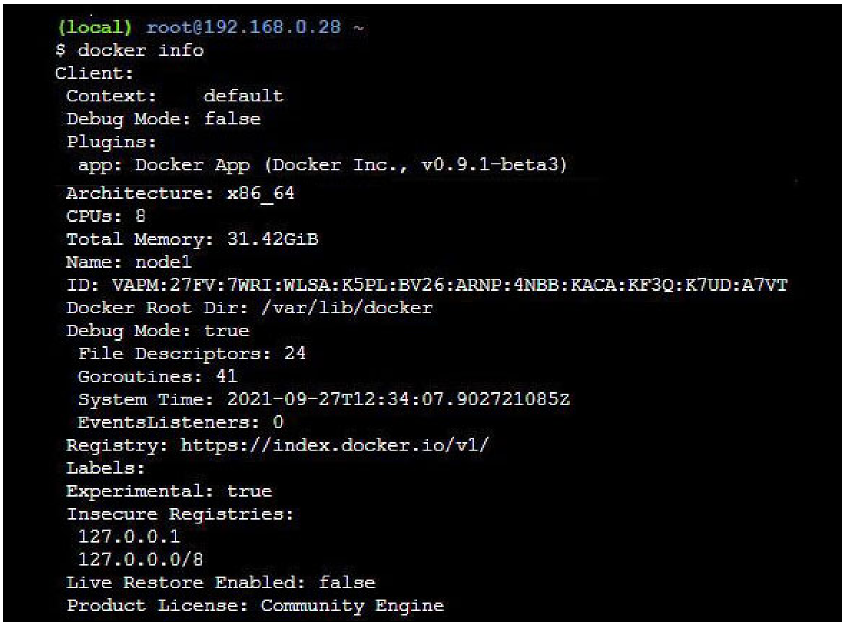


Figure 2: Traditional vs new generation virtualisation using Docker



```
(local) root@192.168.0.28 ~
$ docker info
Client:
 Context:    default
 Debug Mode: false
 Plugins:
  app: Docker App (Docker Inc., v0.9.1-beta3)
 Architecture: x86_64
 CPUs: 8
 Total Memory: 31.42GiB
 Name: node1
 ID: VAPM:27FV:7WRI:WLSA:K5PL:BV26:ARNP:4NBB:KACA:KF3Q:K7UD:A7VT
 Docker Root Dir: /var/lib/docker
 Debug Mode: true
  File Descriptors: 24
  Goroutines: 41
  System Time: 2021-09-27T12:34:07.902721085Z
  EventsListeners: 0
 Registry: https://index.docker.io/v1/
 Labels:
 Experimental: true
 Insecure Registries:
  127.0.0.1
  127.0.0.0/8
 Live Restore Enabled: false
 Product License: Community Engine
```

Figure 3: View Docker version and related information

After installation of Docker, the version can be verified using the following instruction:

```
$ docker version
```

The detailed information on the installed Docker flavour can be displayed using the following code:

```
$ docker info
```

Docker is a powerful platform that enables multiple operating systems to be used on it without creating separate virtual machines in virtualisation software. To work on the Ubuntu OS on Windows with Docker, use the following instruction to download the Ubuntu image:

```
$ docker run -it ubuntu bash
```

Docker Hub

Docker Hub is a cloud based public repository and registry service. Numerous images of assorted platforms and software suites are available on it, which can be pulled and used on Docker